

School of Engineering and Computer Science
MSc Computer Science
Artificial Intelligence and Robotics

Module: MSc CS Project

Detailed Project Proposal

**Driver Distraction Detection System Using
Image Classification**



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1.0 Project Introduction

Driver distraction is a road-safety problem where a driver loses attention because of activities such as phone use, drinking, operating controls, reaching for objects, personal grooming, or interacting with passengers. This project is in the field of artificial intelligence, embedded vision, and embedded vision and hardware supported system design. The problem addressed by the project is how to detect distracted driver behaviour from driver-facing images and use hardware-supported processing to support an alert prototype.

The reason for choosing this topic is that it allows me to apply computer vision and artificial intelligence techniques within an embedded hardware environment, while also developing practical skills in FPGA-based system design. The project allows me to work with a labelled image dataset, develop an image classification approach, use suitable FPGA-related design tools, and demonstrate an electronic alert output. The proposal explains the aim, requirements, research approach, resources, risks, plan, objectives, and expected outcomes.

2.0 Project Goal or Aim

The aim of this project is to design and evaluate a driver distraction detection prototype with hardware supported implementation that uses image data to classify driver behaviour and provide an alert when distracted driving behaviour is identified. The project will focus on dataset preparation, image-based driver behaviour classification, hardware supported or FPGA related design, and prototype evaluation.

3.0 Project requirements

3.1 Core project requirements

The core requirements of this project define the main work that will be carried out to achieve the project aim. The project will focus on dataset preparation, driver behaviour classification, FPGA-related design, and alert output testing.

- To collect, study, and prepare the selected driver distraction image dataset for use in the project. This will include checking the dataset structure, class labels, image filenames, driver subject information, and class distribution.
- To organise the dataset into suitable training, validation, and testing sets. As the dataset contains driver subject information, the data will be prepared carefully to support a fairer evaluation of the system.
- To develop and test an image-based driver behaviour classification approach that can classify driver-facing images into the available dataset classes, such as safe driving, phone use, texting, drinking, operating controls, reaching behind, hair and makeup, or talking to a passenger.
- To apply suitable image preprocessing before classification. This may include image resizing, format conversion, normalisation, and preparation of image input for training, testing, and prototype evaluation.
- To design and verify suitable FPGA-related hardware components where appropriate. The FPGA-related part may support image preprocessing, control logic, decision handling, data flow, or alert output control, depending on the final implementation path and available hardware.
- To integrate the classification output with a decision-making stage and demonstrate an electronic alert output, such as an LED, buzzer, or display indicator, when distracted driver behaviour is detected.
- To evaluate the prototype using suitable testing evidence, including classification results, class-level performance, hardware simulation or implementation results, and alert output behaviour.

3.2 Advanced Project aims

The advanced aims are intended to improve the quality of the investigation and provide a stronger analysis of the developed system. These aims will be considered after the core project requirements have been addressed.

- To compare the project results with relevant research in driver distraction detection and embedded vision systems.
- To analyse which driver behaviour classes are detected correctly and which classes are more likely to be confused.
- To test the prototype under practical conditions where possible, such as using different image inputs, lighting conditions, or camera positions, and identify possible improvements for future development.

4.0 Secondary Research Aims (Literature Review)

The literature review will be used to understand existing work in driver distraction detection, computer vision-based driver monitoring, embedded artificial intelligence, and FPGA-supported image processing. It will help justify the final technical choices and provide a basis for comparing the project results with related research.

- Review research on driver distraction and driver monitoring systems.
- Identify image processing and classification methods used for driver behaviour recognition.
- Review FPGA-based image processing and hardware-supported vision systems.
- Identify suitable evaluation methods for classification, hardware testing, and prototype behaviour.

5.0 Primary Research

Phase	Planned work
Dataset preparation	Study the dataset files, map the classes, organise image paths, and prepare training and evaluation data.
Classification development	Train and test an image-based driver behaviour classification approach using the prepared dataset.
FPGA-related development	Use suitable FPGA development tools to design and test hardware-supported parts such as preprocessing support, data control, decision logic, or alert logic.
Prototype integration	Connect the classification decision to electronic alert outputs and verify the system flow.
Evaluation	Analyse classification results, hardware evidence, alert behaviour, limitations and improvements.

6.0 Project resources

Resource type	Resources
Dataset	State Farm Distracted Driver Detection dataset and associated CSV files. [2]
Software	Python environment, image processing libraries, machine learning framework, FPGA design software, hardware description language, and simulation tools.
Hardware	FPGA development board, camera or image input method, LED, buzzer, and basic electronic components. [3]
Documentation	Dataset documentation, research papers, FPGA design documentation, and project supervision notes. [4] [5]

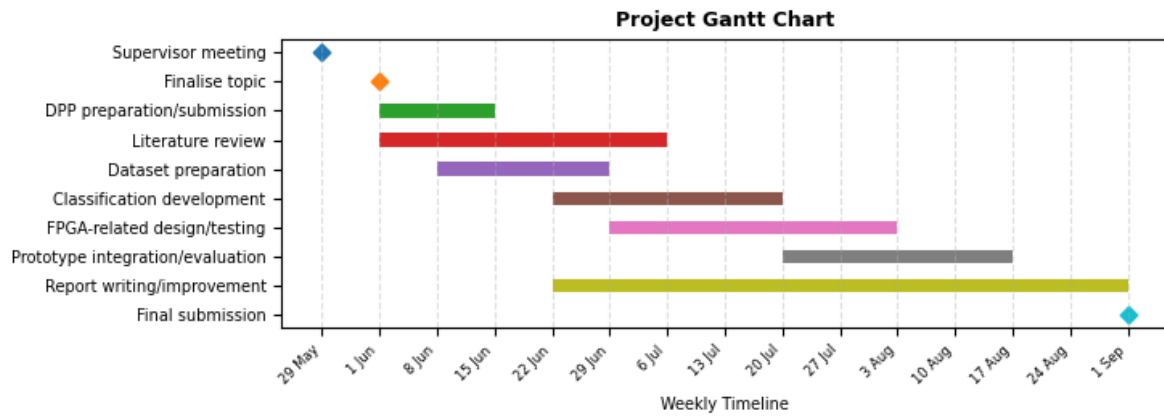
The selected dataset for this project is the State Farm Distracted Driver Detection dataset. It contains labelled driver-facing images for an image classification task. The associated training information file contains 22,424 labelled image records and driver subject information, while the submission-format file contains 79,726 test-image rows. The dataset includes ten behaviour classes such as safe driving, texting, phone use, operating vehicle controls, drinking, reaching behind, hair and makeup, and talking to a passenger. The subject information will be considered during data preparation to reduce unfair testing caused by using the same driver in both training and evaluation data.

7.0 Project risks and their mitigation

Risk description	Probability	Possible effect	Mitigation method
FPGA implementation difficulty	Medium	Hardware progress may slow down	Keep the hardware scope flexible and use simulation evidence where needed.
Dataset preparation issue	Low	Training may be delayed	Prepare and verify the dataset early.
Classification result limitation	Medium	Results may need improvement	Review preprocessing and compare suitable approaches.

Hardware availability issue	Medium	Prototype may be affected	Use available hardware and maintain a simulation pathway.
Time limitation	Medium	Some optional features may be reduced	Prioritise the two approved objectives.

8.0 Project Plan and Objectives



Objective 1	To prepare and use a suitable driver distraction image dataset for training, testing and evaluating a driver behaviour classification system.
Objective 2	To develop and evaluate a driver distraction detection prototype using hardware supported design and electronic alert outputs, with FPGA related implementation considered where appropriate.

9.0 Project outcomes and lessons to be learned

- A documented dataset preparation workflow for driver distraction image classification.
- A trained and evaluated driver behaviour classification system.
- An FPGA-related hardware design or hardware-supported implementation with evidence from testing.
- An electronic alert prototype connected to the detection decision.
- A final report explaining design choices, implementation, evaluation, limitations, and future improvements.

The project will develop practical knowledge in computer vision, dataset handling, hardware description, embedded hardware supported systems, electronics integration, and safety-related embedded AI applications.

References

1. AMD (2024) *Vivado Design Suite User Guide: Design Flows Overview*. Available at: <https://docs.amd.com/r/en-US/ug892-vivado-design-flows-overview> (Accessed: 5 June 2026).
2. Kaggle (2016) *State Farm Distracted Driver Detection*. Available at: <https://www.kaggle.com/competitions/state-farm-distracted-driver-detection> (Accessed: 5 June 2026).
3. Li, Z., Hong, F.Z. and Yue, C.P. (2024) 'FPGA-based acceleration of neural network for image classification using Vitis AI', *arXiv*. Available at: <https://arxiv.org/abs/2412.20974> (Accessed: 5 June 2026).
4. Mofid, N., Bayrooti, J. and Ravi, S. (2020) 'Keep your AI-es on the road: tackling distracted driver detection with convolutional neural networks and targeted data augmentation', *arXiv*. Available at: <https://arxiv.org/abs/2006.10955> (Accessed: 5 June 2026).
5. National Highway Traffic Safety Administration (2024) *Distracted Driving*. Available at: <https://www.nhtsa.gov/risky-driving/distracted-driving> (Accessed: 5 June 2026).